

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2022-23

Paper No. CHB-601: Analytical Chemistry-II

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer five questions in total, including Question 1 which is compulsory.

Question 1. Answer all parts of the following: [5 × 2 = 10]

- (a) List various laboratory neutron sources. How is sensitivity affected by the neutron flux of the source?
- (b) Explain the term: reverse-phase chromatography.
- (c) Differentiate between cation exchanger and anion exchanger resins.
- (d) Write the relationship between radioactive half-life ($t_{1/2}$) and decay constant (λ).
- (e) What is the role of a monochromator in a spectrophotometer?

Question 2.

- (a) A solution of organic acid has distribution coefficient $K_D = 10.0$ between ether and water. If 10.0 g of acid is dissolved in 1.0 L of water, compare the amounts of acid extracted by: (i) single extraction with 1.0 L of ether, and (ii) two successive extractions with 500 mL portions of ether. [8]
- (b) Discuss the isotope dilution method applied in radiochemical analysis. Mention its typical applications. [6]

Question 3.

- (a) Discuss the principles and applications of Neutron Activation Analysis (NAA). [6]
- (b) Calculate the percentage distribution of a substance after three transfers in a Craig counter-current apparatus if the volume of the stationary phase is 4.0 mL, the volume of mobile phase is 8.0 mL, and the distribution coefficient (K_D) is 6.0. [8]

Question 4.

- (a) State and derive Beer-Lambert's law. Describe and compare different reasons for deviations from Beer's law (chemical, instrumental, and real deviations). [8]
- (b) Potassium dichromate ($K_2Cr_2O_7$) aqueous solution does not obey Beer's law upon dilution. Comment and suggest how you would make this solution obey Beer's law. [6]

Question 5. Answer any four of the following:

[4 × 3.5 = 14]

- (a) Describe the procedure to determine the pK_a of an acid-base indicator spectrophotometrically.
- (b) Prove that the minimum relative error in concentration in spectrophotometric measurements occurs when $T = 36.8\%$.
- (c) Discuss paper chromatography and HPLC separation mechanisms.
- (d) Compare adsorption chromatography and partition chromatography.
- (e) Draw the block diagram showing the instrumental design of a double-beam spectrophotometer.